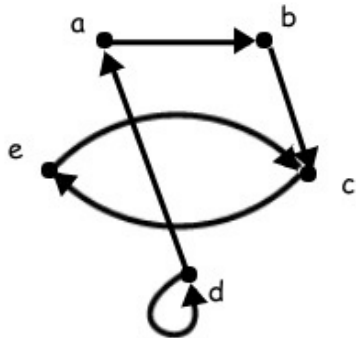


Binary relations on a set:

A binary relation on a set A is a subset of $A \times A$.

Graphical representation of a binary relation on a set:



$$A = \{a, b, c, d, e\}$$

$$R \subseteq A \times A$$

$$R = \{(a, b), (b, c), (e, c), (c, e), (d, a), (d, d)\}$$

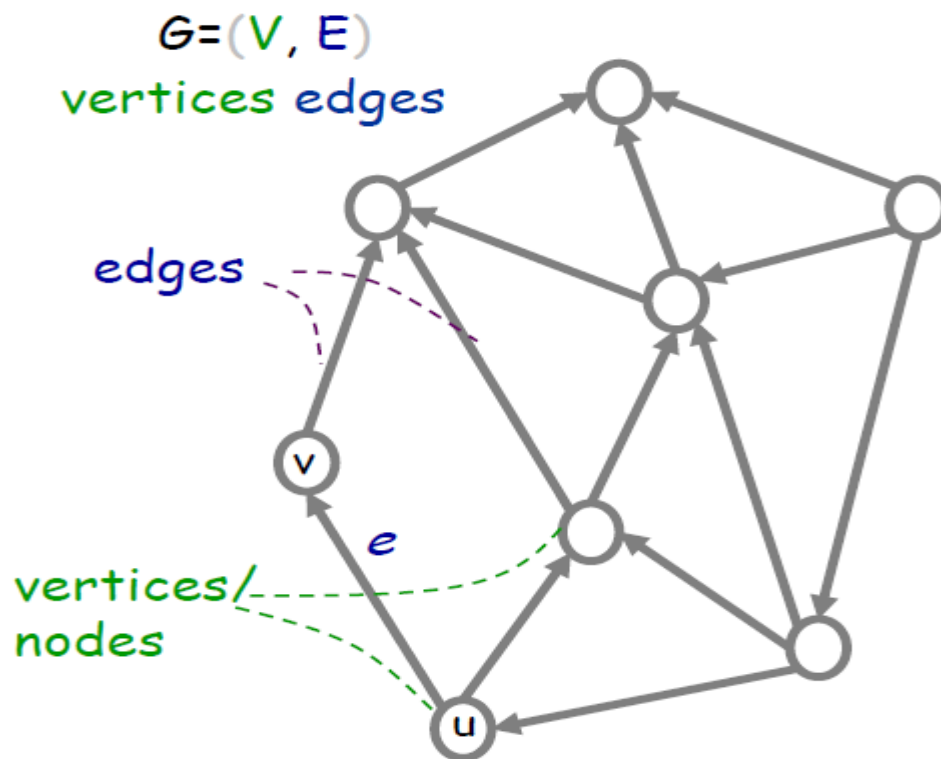
This special case of a binary relation is also called a directed graph.

Directed graphs:

Edge (u, v) goes from vertex u to vertex v .

In-degree of a vertex: the number of edges pointing into it.

Out-degree of a vertex: the number of edges pointing out of it.



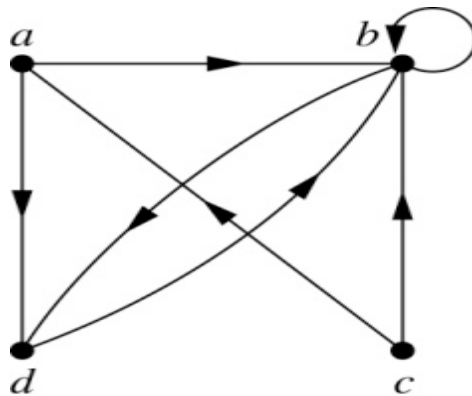
Definition: A directed graph, or digraph, consists of a set V of vertices (or nodes) together with a set E of ordered pairs of elements of V called edges (or arcs).

The vertex a is called the initial vertex of the edge (a, b) , and the vertex b is called the terminal vertex of this edge.

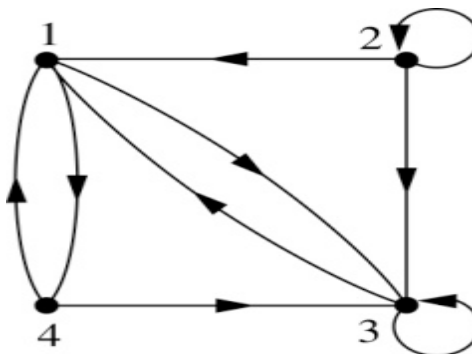
An edge of the form (a, a) is called a loop.

Example 1: A drawing of the directed graph with vertices a, b, c , and d , and edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , and (d, b) .

Solution:



Example 2: What are the ordered pairs in the relation represented by this directed graph?



Solution: The ordered pairs in the relation are $(1, 3)$, $(1, 4)$, $(2, 1)$, $(2, 2)$, $(2, 3)$, $(3, 1)$, $(3, 3)$, $(4, 1)$, and $(4, 3)$